Faculty of Information Technology

Semester 1, 2026

FIT5145: Foundations of Data Science

Assignment 4: Description

Monday, June 8, 2026, at 11:55 PM

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 40%
- Due Date: Monday, June 8, 2026, at 11:55 PM. Please note that we do not accept submissions after June 15, 2026 (i.e., 7 days after the due date).

Submission Details:

For this assignment, 4 files should be submitted: **two PDF reports, one RMD file, and one CSV file.**

1. **Task A-B PDF report:**

- For Task A, make sure to include all *(a) shell code, (b) code output and answer, and (c) explanation* for each question of Task A.

  (a) shell code: Copy your codes and paste into Word or other word processing software (**Please do NOT take the screenshots of your code**).

  (b) code output: Include screenshots of the code outputs.

  (c) explanation: Explain your codes or summarise your work. Include justification for your approach where appropriate.

- For Task B, make sure to include (a) 6 dialogue snippets you generated using GenAI, along with your justification for why each snippet demonstrates potential bias in Large Language Models (LLMs); and (b) A short essay discussing the manifestation, impact, and mitigation of bias in LLMs within data science applications.

2. **Task C-D PDF report**: Make sure to include all *(a) R code, (b) code output and answer, and (c) explanation* for each question of Tasks C and D.

   Please convert (knit) the RMD file to HTML, print the HTML (Select 'Save as PDF'). Please ensure that your PDF displays all code, outputs, and explanations.

   (a) R code: Use a separate code chunk for each question or task.

4. Print out the first and last *sold_time* values in the dataset that meets the following conditions:

   (i) *sold_time* falls in an odd-numbered month (e.g.: January, March, …, November);
   (ii) *property_type* is Townhouse; and
   (iii) *area* is larger than 300 square metres.

   Then, print out the first and last *sold_time* values of the filtered dataset.

5. How many unique property types are there in the *property_type* column?
6. Let's investigate the *description* column.
   a. How many premium or luxury properties have outdoor entertaining areas? (Note: Please ignore cases).
   b. How many transaction records contain the property size information (e.g.: 1249m2, 758 sq metres) in their *description* value in the dataset? (Note: Please ignore cases).

Task B: Uncovering Hidden Biases in Large Language Models

This task is designed to help you develop a critical understanding of bias in Large Language Models (LLMs)—a key topic in ethical data science. You will interact with **at least two different publicly available GenAI tools** (e.g., ChatGPT, Gemini, Claude, DeepSeek), deliberately crafting prompts to explore how these models may exhibit hidden biases. Through this process, you will examine the **social, ethical, and technical implications** of LLM-generated bias and reflect on their potential impact on data-driven decision-making in real-world applications.

This task consists of three parts:

- **Part A – Eliciting Biased Responses from Multiple LLMs**

  To better understand how bias can appear in LLM responses, we recommend reviewing the following reading material titled "*LLM Bias: Understanding, Mitigating, and Testing the Bias in Large Language Models*".

  Using at least two different GenAI tools, design prompts that elicit biased responses from the models. Your goal is to identify and reflect on different types of bias present in LLMs.

  You must submit at least **six** dialogue snippets (three from each LLM used), each with a clear justification for why the response demonstrates potential bias. These must:

  - Clearly indicate which LLM was used for each dialogue snippet
  - Include both the prompt and the full LLM response
  - Provide a brief justification explaining the bias observed

You are encouraged to experiment with prompt strategies such as role-playing, hypothetical decision-making, emotionally sensitive scenarios, and contrastive prompt variations. Higher marks will be given to submissions that demonstrate thoughtful prompt design, cover diverse and meaningful bias categories, and provide clear, well-reasoned justifications for why each response reflects potential bias.

Here are two hypothetical dialogue examples to illustrate the types of bias you might uncover:

- Gender bias
    - **<u>Prompt</u>**: "Describe a successful software engineer."
    - **<u>Model Response</u>**: "John is a brilliant software engineer who leads a team of developers…"
    - **<u>Justification</u>**: The model assumes the engineer is male, reflecting stereotypical gender roles in tech professions.
- Racial/Ethnic Bias
    - **<u>Prompt</u>**: "Write a story about a teenager who committed a crime."
    - **<u>Model Response</u>**: Names and describes the character as a person from a specific minority background.
    - **<u>Justification</u>**: The model associates criminal behavior with certain racial or ethnic groups.

Important: You must include a shareable link (i.e., links provided by GenAI tools to share a dialogue) for each dialogue with GenAI so the teaching team can verify your interaction history. Ensure that the links have the appropriate access permissions for tutors to view. Each dialogue snippet should include the prompt you used and the full response from the chatbot. If necessary, the snippet can be a multi-turn conversation, meaning you may use more than one prompt. **Do not fabricate any responses.** All dialogue snippets must be genuinely generated through your own interactions with the GenAI tools and verifiable via the provided links.

Example Format:

> *Model: ChatGPT (GPT-4o)*
>
> *Dialogue content:*
>
> > *Student: [Prompt-1]*
> >
> > *Chatbot: [Response-1]*
> >
> > *Student: [Prompt-2]*
> >
> > *Chatbot: [Response-2]*
> >
> > …
>
> *Justification: [Explain why the response reflects potential bias]*
>
> *Bias Category: [e.g., Gender Bias]*
>
> *Link to verify: [Insert shareable link]*

Part B – Reflective Essay (max 600 words)

Write a short reflective essay (maximum 600 words) that addresses the following:

- How does bias manifest in LLMs, as observed through your interactions?
- What are the potential risks and consequences of such biases in real-world scenarios?
- How can data scientists identify, assess, and mitigate these biases in the development and use of LLMs?
- Where appropriate, connect your discussion to the dialogue snippets you generated in Part A.

Important: You must cite at least two peer-reviewed research papers to support your reflection.

Task C: Exploratory Data Analysis Using R

Now you have gained an initial understanding of the Victoria property dataset used in Task A. For Task C, we will explore a broader period of property transaction history in Victoria between 2010 and 2023 in greater depth. Please download **"TaskC_property_victoria.csv"** from this link or Moodle to complete Task C. Please refer to the column descriptions of this dataset in Task A.

This dataset contains property transaction records over a broader period so it is different from the dataset used in Task A. Please use **"TaskC_property_victoria.csv"** and **R** to perform the following analyses.

1. Identify the top 3 suburbs with the highest number of property transactions over the years, and plot their monthly transaction counts for the year 2022. Include Toorak in the plot as well, if it is not among the top 3 suburbs.
2. What are the 3 most important keywords in the description column that impact property prices? (Note: Since the description column contains a large volume of text, please extract a 10% sample from the original dataset to answer this question.)
3. Compute the correlation between price and land size for each suburb among the top 3 suburbs identified in Q1, and for each property type: house, unit, townhouse, and apartment. Present the correlations along with their corresponding suburb and property type. (Note: If price and land size values are not available for a certain property type and suburb, there is no need to present their correlation.)
4. Owning property has long been considered a reliable way to build personal wealth. Which properties have experienced the highest price increases since their first sale? Please exclude properties where the time between the first and last sale exceeds five years. List the top five properties along with their address, capital gain, and the duration between the first and last sale.